Chapter 6 Roots: Open Methods

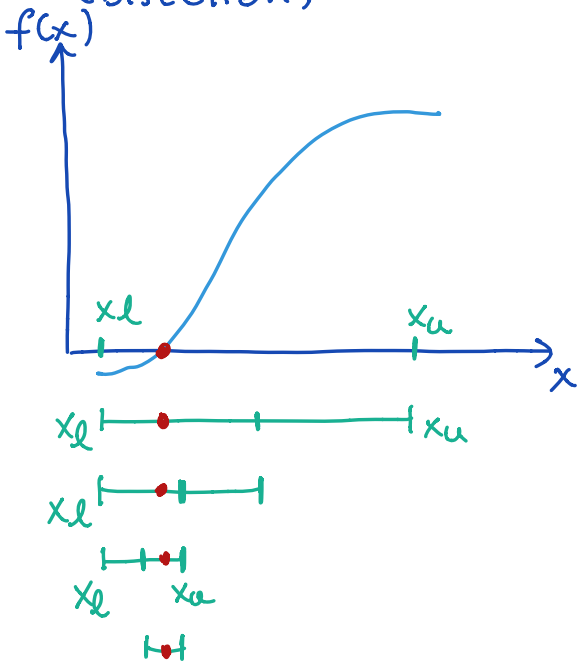
⇒ Bracketing methods are convergent (they move closer to the root at each iteration)

⇒ Open methods:

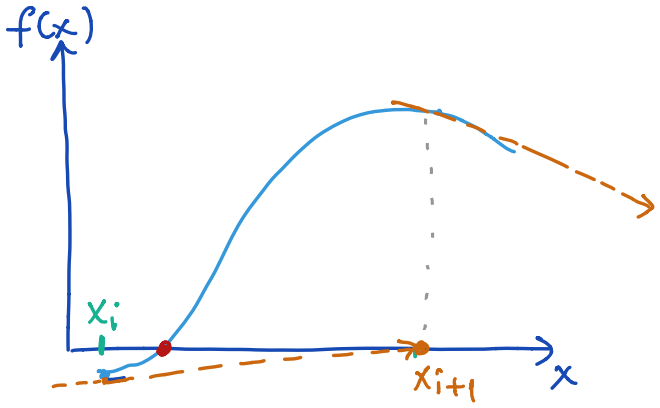
→ Require a single starting point

→ Two starting ptr. that do not necessarily bracket the root.

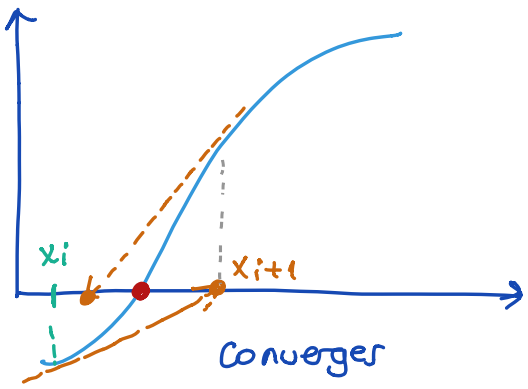
Bracketing Methods (Bisection)



Open Methods



Diverger



Converger

A formula is used to project from x_i to x_{i+1} in an iterative fashion.

Simple Fixed Point Iteration (one-point iteration or successive substitution)

Arrange the function $f(x)=0$ so that x is on the left hand side of the equation (by algebraic manipulation or adding x to both sides)

$$x = g(x) \qquad (1)$$

EX $x^2 - 2x + 3 = 0 \Rightarrow x = \frac{x^2+3}{2}$

$\sin x = 0 \Rightarrow x = \sin x + x$

Eq. (1) provides a way of predicting a new value of x as a function of an old value of x:

$$x_{i+1} = g(x_i)$$

Approximate error: $\varepsilon_a = \left| \frac{x_{i+1} - x_i}{x_{i+1}} \right| 100\,\%$

EX Locate the root of $f(x) = e^{-x} - x$ (True root = 0.56714329)

$$x_{i+1} = e^{-x_i}$$

Starting from $x_0 = 0$

after 10 iterations $x_i = 0.564879$, $\varepsilon_a = 1.11\%$, $\varepsilon_t = 0.399\%$

Alternative Graphical Approach

Solve $f(x)=0$

Separate the equation into two component parts:

$$f_1(x) = f_2(x)$$

Then $y_1 = f_1(x)$ and

$y_2 = f_2(x)$ can be plotted seperately.

The x values corresponding to the intersection of these functions represent the roots of $f(x)=0$

Ex $e^{-x} - x = 0$ Determine the roots graphically.

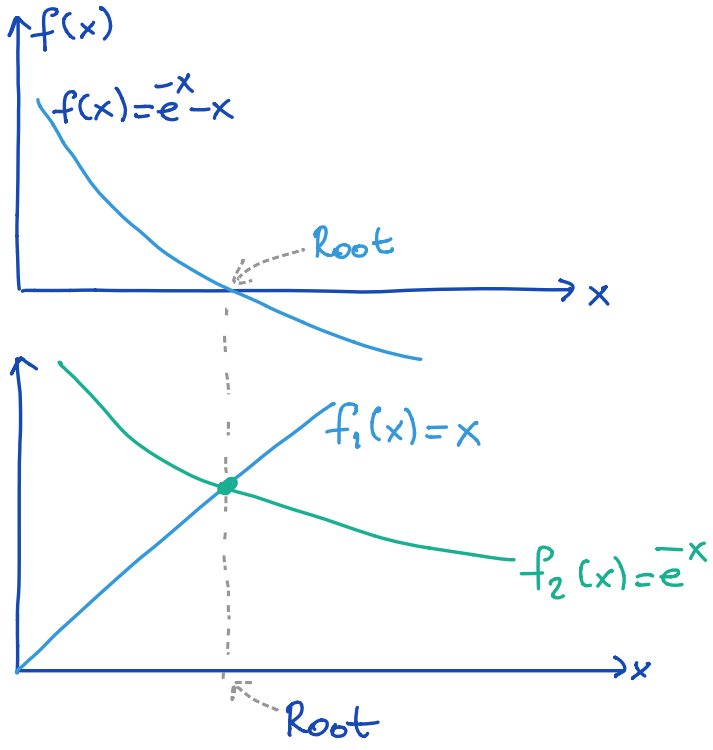
Reformulate as:

$y_1 = x$

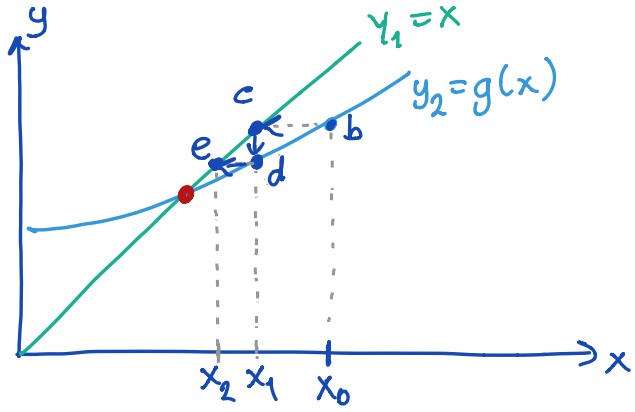
$y_2 = e^{-x}$

$x = e^{-x}$

$y_1 = f_1(x)$ $\quad y_2 = f_2(x)$



Two-curve method to illustrate the convergence and divergence of fixed-point iteration.



convergent

a → initial guer x_0

b → Determine the corresponding point on the y_2 curve $(x_0, g(x_0))$

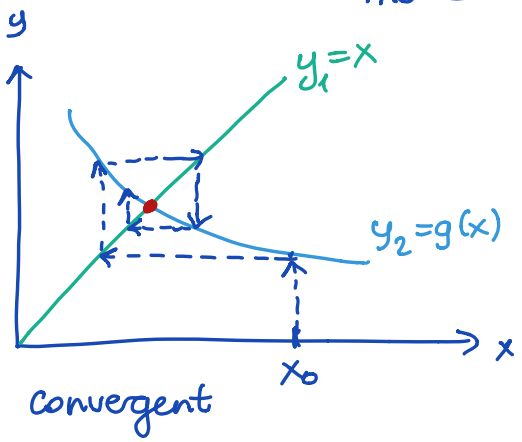
c, d → Locate $(x_1, g(x_1))$ by moving left horizontally to y_1 curve

a, b, c, d are equivalent to the first iteration in the fixed pt. method:

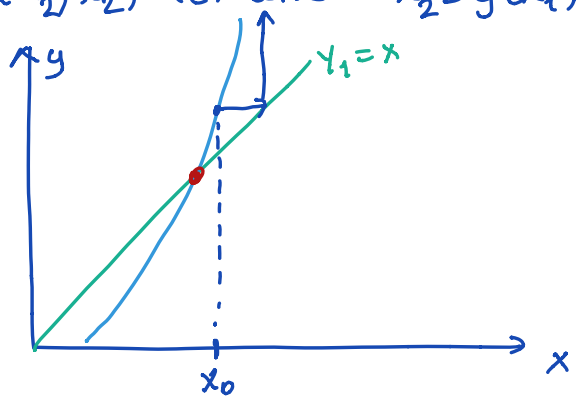
$$x_1 = g(x_0)$$

Next iteration: move to $(x_1, g(x_1))$ (d)

move to (x_2, x_2) (e) where $x_2 = g(x_1)$



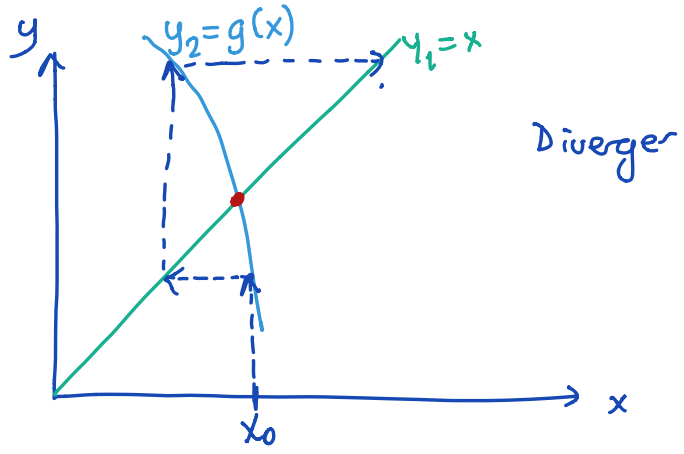
convergent



Diverger

Converges if $|g'(x)| < 1$ around the root

(Derivation is in the book)



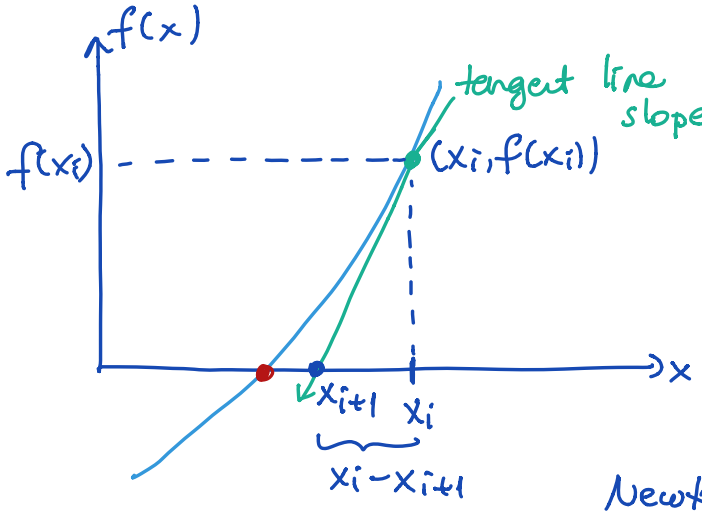
The Newton-Raphson Method

→ Most widely used root-locating method.

→ Initial guess at x_i

→ Extend a tangent from the point $[x_i, f(x_i)]$

→ Point where the tangent crosses the x-axis is an improved root estimate.



Newton-Raphson

Newton-Raphson formula

$$f'(x_i) = \frac{f(x_i) - 0}{x_i - x_{i+1}} \Rightarrow \boxed{x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}}$$

Ex Estimate the root of $e^{-x} - x$, initial guess $x_0 = 0$.

$f'(x) = -e^{-x} - 1$

$$x_{i+1} = x_i - \frac{e^{-x_i} - x_i}{-e^{-x_i} - 1}$$

iteration	x	ε_t (%)
1	0.5	11.8
⋮		
4	0.567143	$< 10^{-8}$

Converger quickly.

Derivation and Error Analysis of Newton-Raphson

Taylor Series:

$$f(x_{i+1}) = f(x_i) + f'(x_i)(x_{i+1} - x_i) + \frac{f''(\xi)}{2!}(x_{i+1} - x_i)^2$$

ξ lies in $[x_i, x_{i+1}]$

$$f(x_{i+1}) \cong f(x_i) + f'(x_i)(x_{i+1} - x_i)$$

at the root $f(x_{i+1}) = 0$

$$0 = f(x_i) + f'(x_i)(x_{i+1} - x_i) \Rightarrow \boxed{x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}}$$

$$E_{t,i+1} = -\frac{f''(x_r)}{2f'(x_r)} E_{t,i}^2$$

$E_{t,i+1}$: true error at iteration (i+1)

x_r: true root

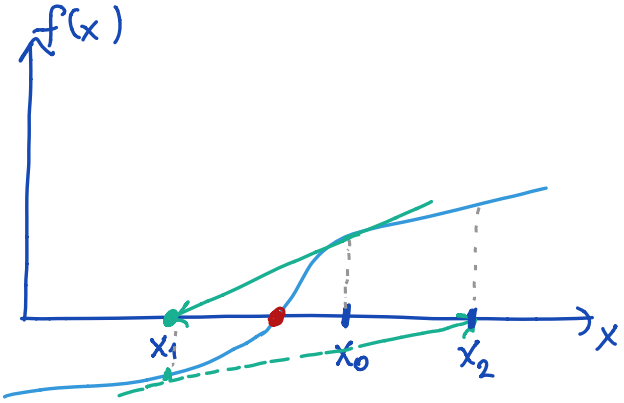
$\Rightarrow$ Error is $\propto$ square of the previous error.

$\Rightarrow$ quadratic convergence.

Problems of Newton-Raphson Method

* There's no general convergence criterion for Newton-Raphson
* Convergence depends on:
  - nature of the function
  - accuracy of the initial guess.

*Remedy: have an initial guess that is "sufficient" close to the root.

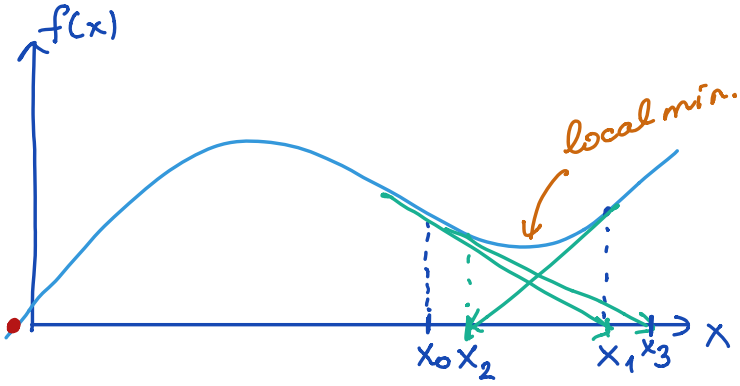


Problem 1: Newton-Raphson

Inflection point $f''(x)=0$ occurs in the vicinity of the root

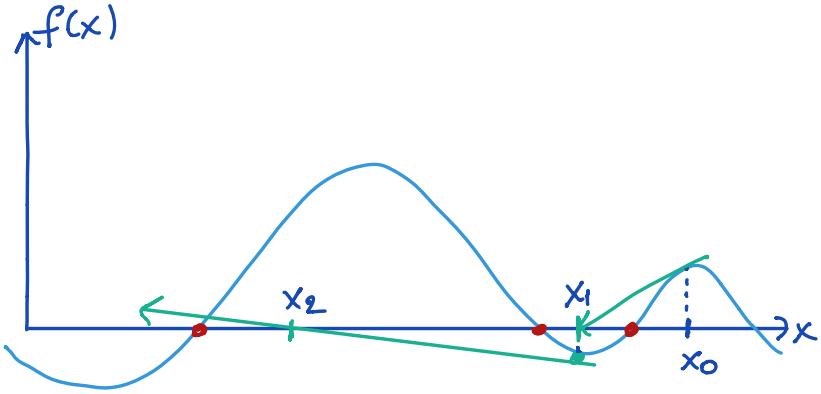
Diverges

Problem 2



Oscillation around a local minimum.

Problem 3



→Initial guess x_0 that is close to one root can jump to a location several roots away (because of near-zero slopes)

→$f'(x_i)=0$ is a disaster since it causes division by zero

in the equation.

Graphically: solution never hits the x-axis.

The Secant Method

→ In Newton-Raphson, the evaluation of the derivative may be a problem.

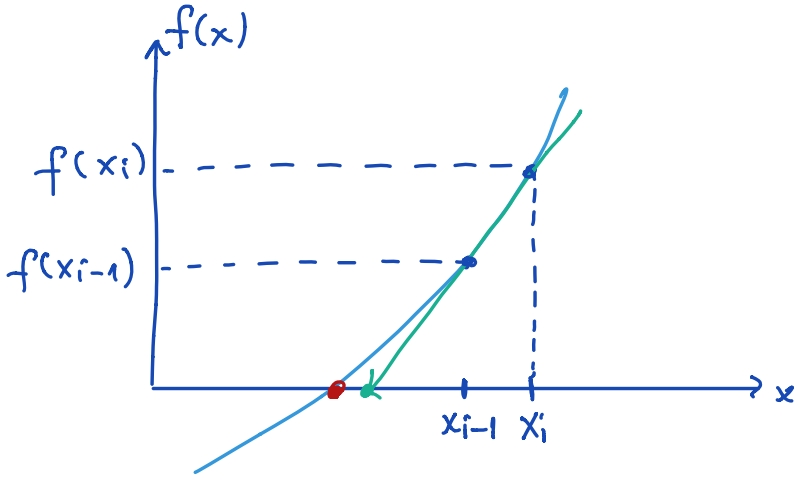
→ Use backward finite difference method for the derivative:

$$f'(x_i) \cong \frac{f(x_{i-1}) - f(x_i)}{x_{i-1} - x_i}$$

↓ Substitute in Newton-Raphson formula

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

$$\boxed{x_{i+1} = x_i - \frac{f(x_i)(x_{i-1} - x_i)}{f(x_{i-1}) - f(x_i)}}$$



Secant Method uses a difference rather than a derivative